FIN 2020 Homework 7 **Due by 23:59 May 5, 2026 (Tuesday)**

Question 1:

Prices of Contingent Claims and Complex Assets:

Consider an economy in which there are two dates $t = 0$ and $t = 1$, and two possible states, $i = 1$ and $i = 2$, at $t = 1$, which occur with equal probability, so that $\pi_1 = \pi_2 = 1/2$. In this economy, two "complex" assets are traded. The first sells for price $P_1^0 = 2$ at $t = 0$ and makes payoffs of $Z_1^1 = 2$ in state $i = 1$ at $t = 1$ and $Z_1^2 = 4$ in state $i = 2$ at $t = 1$. The second sells for $P_2^0 = 1.25$ at $t = 0$ and makes payoffs $Z_2^1 = 1$ in state $i = 1$ at $t = 1$ and $Z_2^2 = 3$ in state $i = 2$ at $t = 1$.

a. Use this information to compute the price q^1 at $t = 0$ of a contingent claim that pays off one in state $i = 1$ at $t = 1$ and zero in state $i = 2$ at $t = 1$ and the price q^2 at $t = 0$ of a contingent claim that pays off zero in state $i = 1$ at $t = 1$ and one in state $i = 2$ at $t = 1$.

b. Use your answers from part (a), above, to compute the price P_b^0 at $t = 0$ of a bond that pays off $Z_b^1 = 1$ in state $i = 1$ at $t = 1$ and $Z_b^2 = 1$ in state $i = 2$ at $t = 1$.

c. Use your answers from part (a), above, to compute the price P_s^0 at $t = 0$ of a share of stock that makes no dividend payments, but can be sold for $P_s^1 = 2$ in state $i = 1$ at $t = 1$ and $P_s^2 = 3$ in state $i = 2$ at $t = 1$.

Question 2:

You observe the following assets with the corresponding state-dependent payoffs:

Securities	A	B	C
States	3	7	8
	1	2	9
	7	16	25

Is this market complete?

If the market is not complete, introduce call options to make the market complete. (Note: The answer is not unique)